6.3.3 Electromagnetism

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	magnetic flux ϕ; the unit weber; $\phi = BA\cos\theta$	
(b)	magnetic flux linkage	
(c)	Faraday's law of electromagnetic induction and Lenz's law	
(d)	**(i)** e.m.f. = − rate of change of magnetic flux linkage; $\mathcal{E} = -\dfrac{\Delta(N\phi)}{\Delta t}$ **(ii)** techniques and procedures used to investigate magnetic flux using search coils	*M3.9*
(e)	simple a.c. generator	
(f)	**(i)** simple laminated iron-cored transformer; $\dfrac{n_s}{n_p} = \dfrac{V_s}{V_p} = \dfrac{I_p}{I_s}$ for an ideal transformer **(ii)** techniques and procedures used to investigate transformers.	*M0.3*